

liquidate underwater positions, similar to the process followed by MakerDAO.

dYdX offers borrowing and lending similar to Compound and Aave. It also features free flash loans (Aave's are not free), which makes it a popular choice for DAI, ETH, and USDC flash liquidity. In the world of open smart contracts, it makes sense that flash loans rates would be driven to zero since they are nearly risk-free. Lending rates are determined by the loan's duration and relative risk of default. For flash loans, repayment is algorithmically enforced, and time is infinitesimal. In a single transaction, only the user can make any function calls or transfers; no other Ethereum users can move funds or make any changes while a particular user's transaction is in flight, resulting in no opportunity cost for the capital. Hence, as expected, a market participant offering free flash loans will attract more usage to their platform. Because flash loans do not require any up-front capital, they democratize access to funds for various use cases. In the Aave example, we show how flash loans can be used to refinance a loan. We will now illustrate the use of flash loans to capitalize on an arbitrage opportunity.

Suppose the effective exchange rate for 1,000 DAI for ETH on Uniswap is 6 ETH/1,000 DAI. (The instantaneous exchange rate would be different due to slippage.) Also, suppose the dYdX DEX has a spot ask price of 5 ETH for 1,000 DAI (i.e., the ETH are much more expensive on